

iPSC Cell Culture – Maintenance and Expansion

General Note: Warm all media before use. Media should be room temperature. Do not use a water bath to warm media.

Matrigel Coating

1. To resuspend, thaw aliquot on ice
2. Add 12.5mL cold DMEM/F12
3. Pipette up and down twice
4. Add 1mL of Matrigel per well of 6 well plate
5. Store diluted Matrigel at 4C

Thawing iPSC

1. Prior to thawing cells, coat plate with Matrigel for 1 hour
 - 1 vial of iPSC should be thawed into 1 well of a 6 well plate
2. Add 9mL DMEM/F12 to a 15mL conical tube labeled with the iPSC line name and passage number
3. Remove cells from liquid nitrogen storage
4. Quickly thaw cells in 37°C water bath and/or in hands
5. Just prior to complete thaw, remove vial from water bath
6. Transfer the contents of the cryo-vial (~1mL) into the 15mL conical tube
7. Spin at 750rpm for 3 minutes at room temperature
8. Aspirate media
9. Resuspend cells in 2mL mTesR1 (supplemented with 5-10µM Rock Inhibitor) by pipetting two times
10. Transfer the cell solution to one well of a 6-well plate
11. Incubate at 37°C overnight in 6% CO₂
12. Replace the media daily until cells are ready to split or analyze

Cell Growth/Maintenance

Note: Media should be changed daily. It is okay to skip a media change one time each week if double feeding is performed; however, this is largely dependent on the density of the cells and volume of media (do not double feed if cells are more than 70% confluent)

1. Aspirate media
2. Gently add fresh mTesR1 to cells (volume depends on cell density and well size)
 - 0.5mL per well to 24 well plate
 - 2-4mL per well to 6 well plate
 - 5-10mL to 10cm² plate
3. Incubate at 37°C in 6% CO₂

Cleaning

Note: When differentiating cells appear in the culture, it is important to remove all the cells promptly. Repeated cleaning may be necessary over the course of several days to remove all the material. If differentiation is excessive and line is precious, perform subcloning.

1. Under microscope, remove differentiated cells with p20 or p200 tip (depending on the amount of differentiation). Transfer the cells/media to a biohazard bag.

2. Gently wash cells with 1x PBS
3. Add fresh mTesR1
 - 0.5mL per well to 24 well plate
 - 2-4mL per well to 6 well plate
 - 5-10mL to 10cm² plate
4. Incubate at 37°C in 6% CO₂ until cells are 60-80% confluent. Change mTesR1 media daily until cells are needed. Repeat cleaning as necessary.

Splitting/Expanding

Note: iPSCs grow on Matrigel. Plates should be coated with Matrigel at least 1 hour prior to plating and no longer than 24 hours prior to plating cells:

- 0.5mL in 12 well plate
- 1mL in 6 well plate
- 4mL in 10cm² plate

It is critical to keep Matrigel on ice while coating. Prior to plating cells, ensure Matrigel has not evaporated from well.

1. Aspirate media
2. Gently wash cells with 1x PBS (2-3 mL/well)
3. Add Accutase (Gibco A11105-01) directly to the cells and incubate at 37°C for 3-4 minutes.
 - 6 well plate, add 0.75-1mL per well
 - 24 well plate, add 0.5mL
 - 10cm² dish, add 3mL
4. Tap dish to aid in dislocation of cells
5. Add DMEM/F12 directly to cells and scrape gently to remove all cells (use p1000 for 24 well plate, and cell scraper for 6 well plate and 10cm² dish)
 - 6 well plate, add 2-4 mL per well
 - 24 well plate, add 1mL
 - 10cm² dish, add 9mL
6. Collect cells in conical tube (15mL/50mL depending on volume)
7. If necessary, add 2-5mL DMEM/F12 to dish to remove all cells from the dish and add to conical tube
8. Centrifuge cells at 750rpm for 3 minutes at room temperature
9. Carefully aspirate supernatant
 - To avoid aspirating cell pellet, it is OK to leave a small amount of media (0.5-1mL)
10. Resuspend cell pellet with mTesR1 (Rock Inhibitor addition varies, see below)
 - 2mL mTesR1 per well of a 6 well plate
 - Our goal is to maintain iPSC lines without using Rock Inhibitor; however, this must be done through careful weaning off Rock Inhibitor
 - All cells should be thawed in Rock Inhibitor:
 - 10µM concentration for new iPSC lines, lines thawed from 96 well after editing.
 - 5µM concentration if thawing from a line without knowledge of its Rock sensitivity.
 - 1µM concentration for all other lines (for lines still exposed to Rock Inhibitor, use 1µM. Otherwise, do not use Rock Inhibitor)
11. Pipet cells 2 times only to preserve clumps
12. Transfer cell suspension to appropriate plate (pre-coated with Matrigel for at least 1 hour)

- For maintenance, dilute cells 1:3 in mTesR1
 - For expansion, plate all cells
13. Incubate at 37°C in 6% CO₂ until cells are 60-80% confluent. Change mTesR1 media daily until cells are needed

iPSC Freezing

1. Aspirate media
2. Gently wash cells with 1x PBS
 - Use 2-3mL per well in 6 well plate
3. Add Accutase (Gibbco A11105-01) directly to the cells and incubate at 37°C for 3-4 minutes
 - 6 well plate, add 0.75-1mL per well
 - 10cm² dish, add 3mL
4. Tap dish to aid in dislocation of cells
5. Add DMEM/F12 directly to cells
 - 6 well plate, add 2-4mL per well
 - 10cm² dish, add 9mL
 - If cells remain attached, use a cell scraper to gently dislodge cells (apply gentle pressure and use 1-2 passes to remove cells)
6. Collect cells in conical tube (15mL/50mL depending on volume)
7. Add 2-5mL DMEM/F12 to dish to remove all cells from the dish and add to conical tube
8. Centrifuge cells at 750rpm for 3 minutes at room temperature
9. Carefully aspirate supernatant
 - To avoid aspirating cell pellet, it is OK to leave a small amount of media (0.5-1mL)
10. Resuspend cell pellet with mTesR1 (No Rock Inhibitor)
 - Use volume appropriate for freezing
 - Assume 1mL per cryovial total and add ½ total volume of mTesR1
 - Pipet cells 1-2 times only to preserve cell clumps
 - **Example:** to freeze 10 tubes, you will need 10mL total and will add 5mL mTesR1 to cell pellet (and 5mL of 2x Freezing Media below)
11. Add an equal volume of cold 2x Freezing Media (20% DMSO, FBS). Pipet cells 1 time only to preserve cell clumps
12. Transfer cell suspension to pre-labeled cryovials (1mL per cryovial)
 - Ensure that cryovials are labeled with the following
 - Cell Type
 - Line Name
 - Passage #
 - Date
 - Your Name
13. Freeze vials at -80°C in foam racks for 48-72 hours
14. Transfer vials to liquid nitrogen for long-term storage